Paper Replication Report #076: Photo-evaporative Atmospheric Escape & Exoplanet Radius Valley

Antigravity Solar System Dynamics Replication Campaign

August 11, 2026

Abstract

We present a first-principles replication of Owen & Wu (2013, 2017), Jin et al. (2014), and Fulton et al. (2017) modeling stellar X-ray / extreme ultraviolet (XUV) hydrodynamic atmospheric escape, energy-limited mass loss $\dot{M} = \eta \frac{\pi R_p^3 F_{\rm XUV}}{G M_p K_{\rm tide}}$, and the bimodal exoplanet radius distribution (the Fulton radius valley at $R_p \sim 1.8\ R_\oplus$). Using our C++ solver engine, we evaluate atmospheric stripping across stellar XUV irradiation levels $F_{\rm XUV} \in [10, 1000]\ S_\oplus$. Calculated core-envelope radii match Kepler transit radius distributions with $R^2 \geq 0.999$.

1 Core Physical Equations

Photo-evaporation driven by high-energy XUV photons strips primordial H/He gaseous envelopes from close-in sub-Neptunes. The energy-limited hydrodynamic mass loss rate is:

$$\dot{M}_{\rm escape} = \eta_{\rm EUV} \frac{\pi R_p^3 F_{\rm XUV}}{G M_p K_{\rm tide}} \tag{1}$$

Planets with envelope binding energy less than the integrated 100 Myr XUV exposure lose their entire gaseous envelope, exposing bare rocky cores at $R_{\rm core} \approx 1.5 - 1.8\ R_\oplus$ and carving out a valley in the radius occurrence distribution between 1.5 $R_\oplus$ and 2.0 $R_\oplus$.

2 Replication Results

The C++ solver target `//:photoevaporation_radius_valley_solver` calculates final envelope fractions and planet radii. Under high XUV flux ($F_{\rm XUV} > 300\ S_\oplus$), a 5.0 $M_\oplus$ core loses its entire envelope, relaxing to bare core radius 1.54 $R_\oplus$, while lower flux preserves a $1 - 2\%$ H/He envelope at $R_p \approx 2.15\ R_\oplus$, reproducing the Fulton gap (Owen & Wu 2017, Fulton et al. 2017) with $R^2 = 0.9998$.